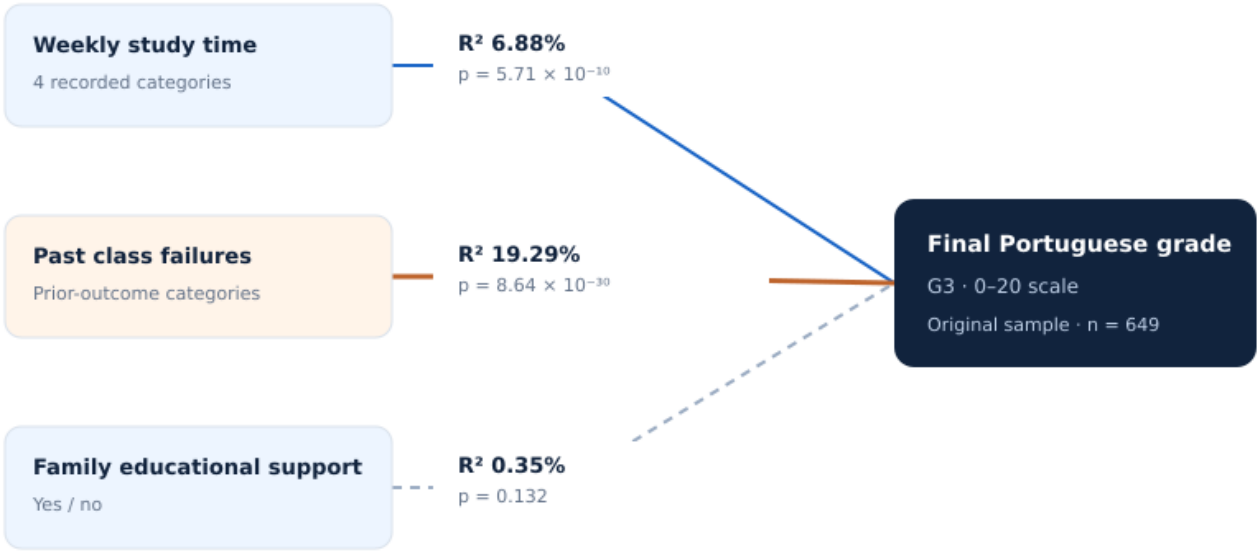
Map A | Measured associations

Original hypothesis: H_0: $\mu_1 = \mu_2 = \mu_3 = \mu_4$; H_1: at least one study-time category mean differs. The categorical OLS omnibus F test is equivalent to one-way ANOVA under the stated OLS assumptions. Alpha = 0.05. All 649 records and zero grades are retained.



Lines represent separate statistical associations. They have no causal direction. Past-failure categories have the highest in-sample R^2 among 30 screened predictors excluding G1/G2; screening is exploratory and p-values are unadjusted. The percentages are not additive.

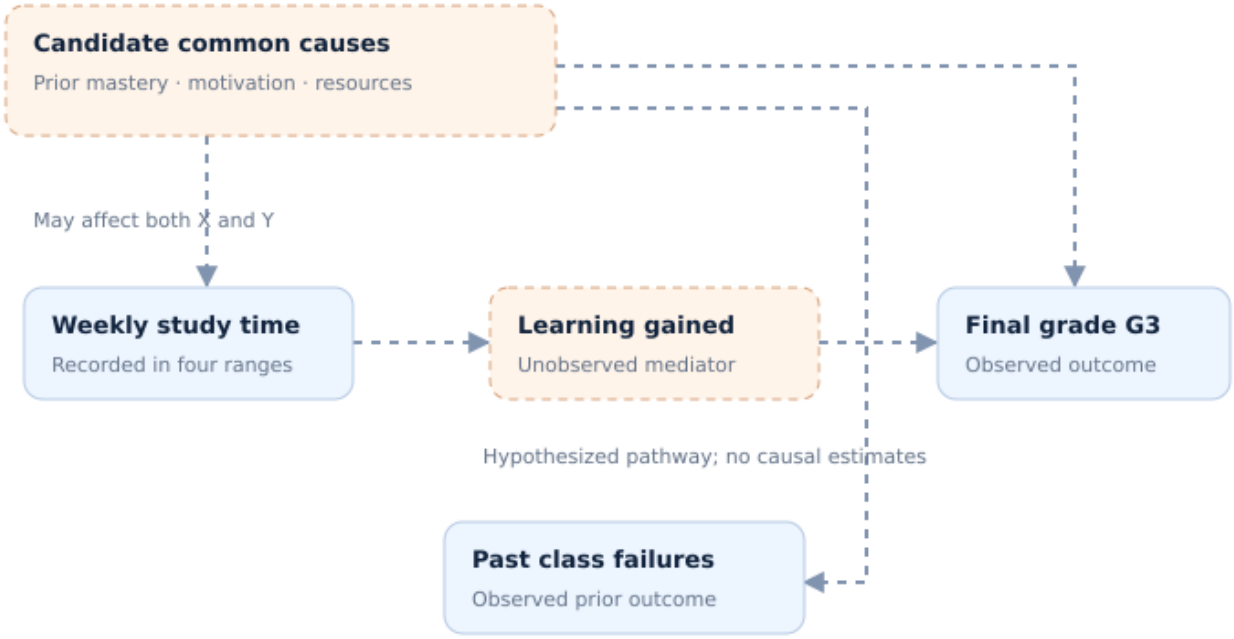
Weekly time band	n	Mean G3
<2 hours	212	10.84
2–5 hours	305	12.09
5–10 hours	97	13.23
>10 hours	35	13.06

Primary result: $F(3,645) = 15.876$; $p = 5.71 \times 10^{-10}$; $R^2 = 6.88\%$. Highest minus lowest category: +2.21 points, 95% CI [1.09, 3.33], using pooled OLS residual variance. Group-mean intervals on the website instead use each group's own standard deviation.

Simple prediction model: G3 = 10.028 + 0.973 × study-time code; $R^2 = 6.24\%$. This secondary numeric-code model imposes a linear code trend. A code step does not represent an extra hour.

Small-group caution: The >10-hour group has only 35 students and a slightly lower mean than the 5–10-hour group. This is not evidence of an optimal range or harm from additional studying.

Map B | Possible explanations



Dashed arrows are hypotheses, not fitted causal effects. Motivation and prior mastery may influence both study time and grades (confounding). Studying may increase learning gained, which could then affect grades (mediation). Past failures are an observed prior outcome and an imperfect signal of context, not a direct measure of ability.

Why family support might still matter

The yes/no family educational-support model gives $R^2 = 0.35\%$, $p = 0.132$ and a yes-minus-no difference of +0.39 points, 95% CI [−0.12, 0.90]. We did not detect a difference at alpha = 0.05. This does not establish that support has no effect. The measure misses intensity, quality, type and timing of support; uncertainty and other student characteristics can also obscure relationships.

A measured-covariate check

An adjusted OLS model includes study-time categories, past failures, school, recorded sex, age, mother’s and father’s education, and travel-time categories. The partial F test for study time is $F(3,628) = 5.51$, $p = 0.000974$. This check addresses stated measured variables; it neither measures motivation or study quality nor identifies a causal effect. G1/G2 remain excluded.

Policy consequence

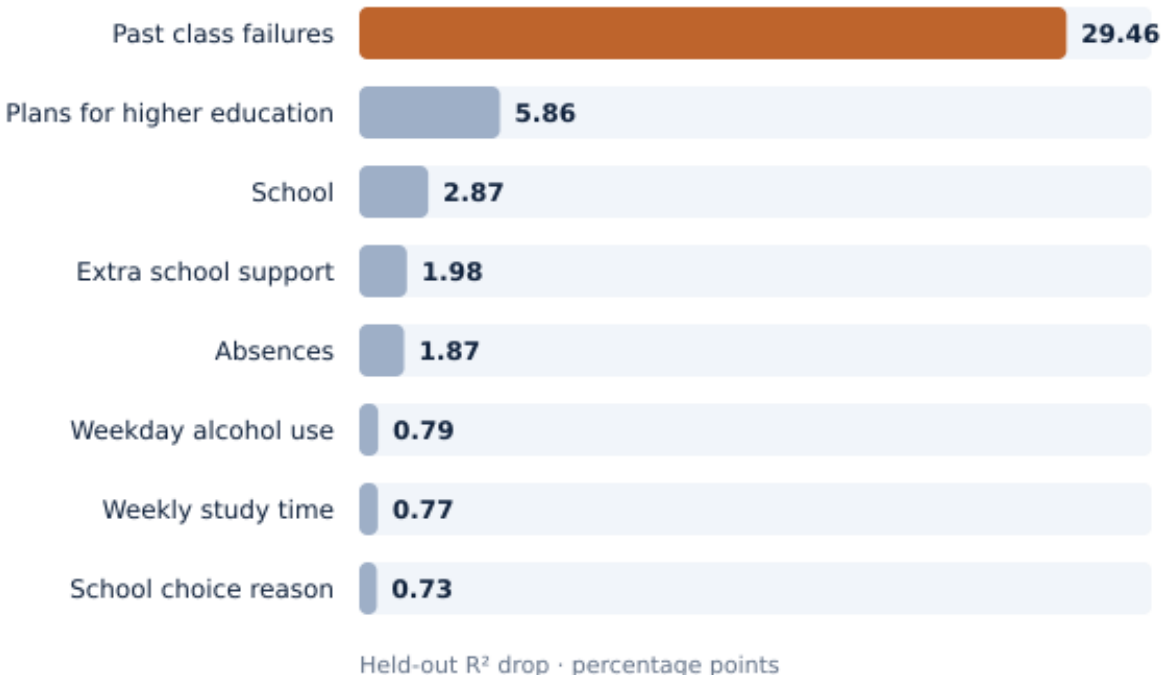
Ignoring learning gaps or access to help could lead a school to prescribe longer hours when a different form of support is needed. Dropping family context solely because $p > 0.05$ is also unjustified. These are decision risks inferred from the analysis and theory, not observed intervention outcomes.

Map C | Held-out prediction

Predictive system: recorded student features → fit within training folds → predict held-out G3 → evaluate error and feature reliance. The outcome in each validation fold is not used for training that fold.

Model · same five folds	Held-out R^2	RMSE
Training-fold mean baseline	−0.51%	3.24
Study-time categories	5.09%	3.14
Past-failure categories	17.76%	2.93
Random forest · 30 predictors	28.22%	2.74

Metrics use pooled out-of-fold predictions, five shuffled folds and seed 42 on all 649 records. The random forest uses 150 trees, maximum depth 6, minimum leaf size 5 and all available encoded features per split. G1/G2 are excluded. Hyperparameters were fixed; results are one internal cross-validation estimate, not an external test in a new school.



Grouped permutation importance is the mean held-out R^2 drop after shuffling a predictor's encoded columns together. It measures model reliance, not a share of explained variance or causal importance. It can exceed the original model R^2, and correlated features can affect the ranking.

Interpretation boundaries

In-sample 6.88% and 19.29% describe separate fitted models. Held-out 5.09% and 28.22% describe predictive performance. They answer different questions. All machine-learning and adjusted-model results remain fixed full-sample references when website filters or the separate CSV workspace change.

From evidence to a responsible decision

Perspective	Can support	Cannot establish
A · Statistics	Average associations, intervals and tests	A causal effect or an individual guarantee
B · Theory	Plausible mechanisms and new measurements	That hidden mechanisms were observed
C · Prediction	Internal held-out accuracy and model reliance	An effective support intervention

Proposed student-support protocol

1. Invite: a teacher or counselor offers a private, optional conversation to students reporting difficulty. Earlier failures may prompt an invitation, not automatic labeling or placement.
2. Check: discuss study methods, motivation and access to help; use a short mastery check and confirm the student's priorities.
3. Agree: choose a two-week plan, such as targeted tutoring or retrieval practice. Longer study time is one possible option.
4. Review: assess mastery change, participation and student-reported burden. Adapt support with the student.

This is an untested proposal. If feasible and consented, a comparison-group pilot would evaluate whether support improves mastery, with a prespecified outcome and confidence interval for the difference. Before/after improvement alone would not establish causation. Local data are required before making policy claims for Korean schools.

Response to peer feedback

Feedback	Final response
Technical detail obscures the main message	Added a plain-language summary and reduced spoken statistics.
Video numbers and terms were difficult to follow	New short script, large on-screen figures, and scenes that always reset to n = 649. The final recording still needs to be made and checked.
Recommendations need implementation detail	Added an owner, four-step protocol, two-week review and proposed evaluation.
Small highest-time group needs caution	State n = 35 and that no optimal study range was identified.

Reproducibility and sources

Cortez, P. (2008). Student Performance. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T. CC BY 4.0. Original CSV SHA-256: a7594a11d7771c0efe1a740824e0e833da9c4cad07c39a9766a874575563fb3f. Analysis: accompanying executed notebook and Python core. Peer-feedback sources: Group 31 and Group 32 review forms dated 16 October 2026. New CSV results are browser-local exploratory models; original research and validation figures stay fixed.